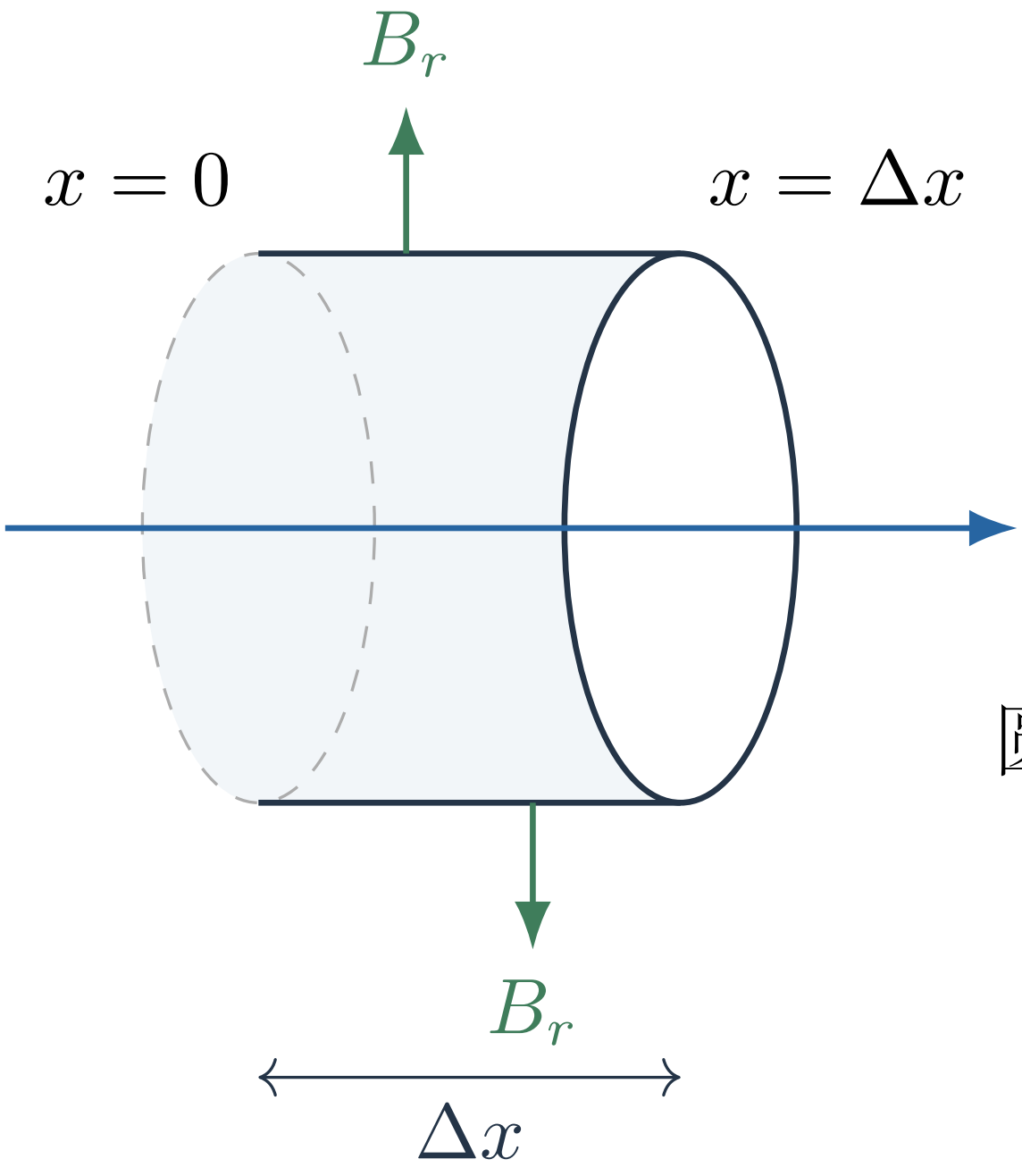

侧面积 $2\pi R\Delta x$

$\Delta\Psi_{\text{ext}} = G\Delta x$

圆面正法向取 $-x$